
Microeconomics I

Class 4 — Utility maximisation

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Short quiz

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top of the sheet.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

Plan for today

10:00	Short quiz
10:08	Practicalities; Ann and Barbara
10:14	Budget sets
10:26	The consumer's problem; Walrasian demand
10:50	Break
11:00	Kuhn–Tucker: tangency and corners
11:35	CES demand; summary

Practicalities

- Problem set 1: due at the start of class Wed 14 Oct, on paper, one copy per group
- Office hours: Wednesdays 12:00–13:00, after class
up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Ann or Barbara: who is better off?

Same college, similar jobs, both started at \$30 000.

Ann

first year: no inflation

raise 2 %

Barbara

first year: 4 % inflation

raise 5 %

Who is doing better in economic terms? Who is happier?

Shafir, Diamond & Tversky (1997), Quarterly Journal of Economics 112(2), Problem 1.

Most say Ann is better off, and Barbara happier

better off in economic terms

71 %

said Ann

happier

64 %

said Barbara

Judged by the number on the payslip, not by what it buys: **money illusion**.

Back to it on slide 15.

Shafir, Diamond & Tversky (1997), "Money Illusion", QJE 112(2): 341–374, Problem 1; three separate groups of US respondents.

Budget sets

MWG 2.D

Definition: the Walrasian budget set

$$\mathcal{B}_{p,w} = \{x \in \mathbb{R}_+^2 : p \cdot x \leq w\}$$

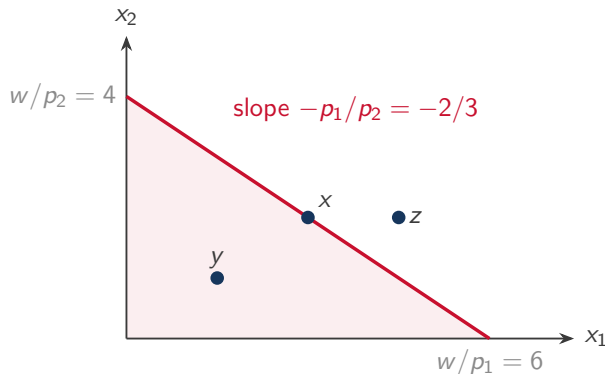
prices $p \gg 0$, wealth $w > 0$

budget line: $p \cdot x = w$

Which of x, y, z are in $\mathcal{B}_{p,w}$?

MWG Definition 2.D.1, Figure 2.D.1.

Here $p = (2, 3)$, $w = 12$.



Definition: the Walrasian budget set

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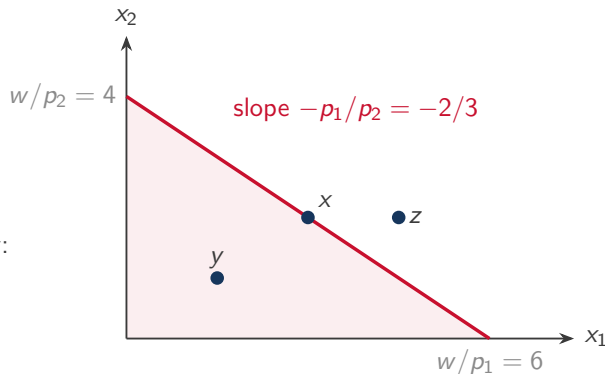
budget line: $p \cdot x = w$

Which of x, y, z are in $\mathcal{B}_{p,w}$?

x costs 12 = w , y costs 6, z costs 15 > w :
 x and y .

MWG Definition 2.D.1, Figure 2.D.1.

Here $p = (2, 3)$, $w = 12$.



The price vector is perpendicular to the budget line

x, x' on the line:

$$p \cdot (x' - x) = w - w = 0$$

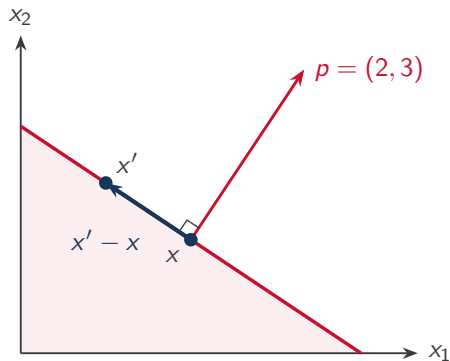
p points out of $\mathcal{B}_{p,w}$, perpendicular to its edge

$$x, x' \in \mathcal{B}_{p,w}, \alpha \in [0, 1] \Rightarrow$$

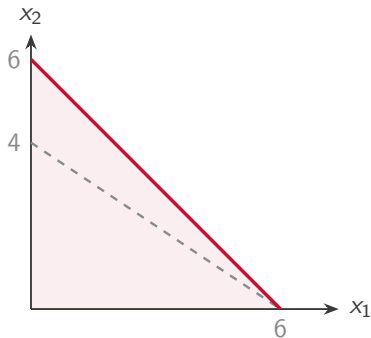
$$\alpha x + (1 - \alpha)x' \in \mathcal{B}_{p,w}$$

the budget set is convex

MWG 2.D, Figure 2.D.3 and p. 22.

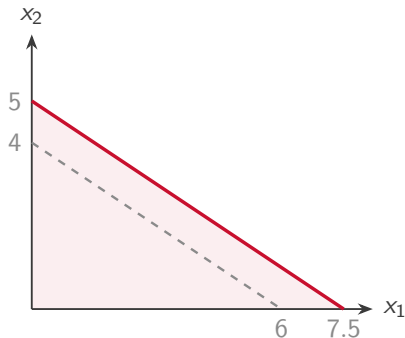


A lower price tilts the line; more wealth shifts it



$p_2: 3 \rightarrow 2$

pivots at (6, 0); slope $-2/3 \rightarrow -1$



$w: 12 \rightarrow 15$

parallel shift; slope stays $-2/3$

MWG Figure 2.D.2 (a fall in p_2).

In pairs: what happens to the budget set?

Start: $p = (2, 3)$, $w = 12$: intercepts 6 and 4, slope $-2/3$.

new (p, w)	intercepts	slope	the new set is ...
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$(3, 4.5), 18$

$(2, 3), 9$

$(1, 3), 12$

$(3, 2), 12$

Four minutes. Intercepts first: w/p_1 and w/p_2 .

In pairs: what happens to the budget set?

Start: $p = (2, 3)$, $w = 12$: intercepts 6 and 4, slope $-2/3$.

new (p, w)	intercepts	slope	the new set is ...
$(3, 4.5), 18$	6 and 4	$-2/3$	the same set
$(2, 3), 9$	4.5 and 3	$-2/3$	smaller: parallel shift in
$(1, 3), 12$	12 and 4	$-1/3$	larger: pivot at $(0, 4)$
$(3, 2), 12$	4 and 6	$-3/2$	neither: they cross at $(2.4, 2.4)$

Four minutes. Intercepts first: w/p_1 and w/p_2 .

The consumer's problem

MWG 3.D

A best affordable bundle exists: continuous u on a compact set

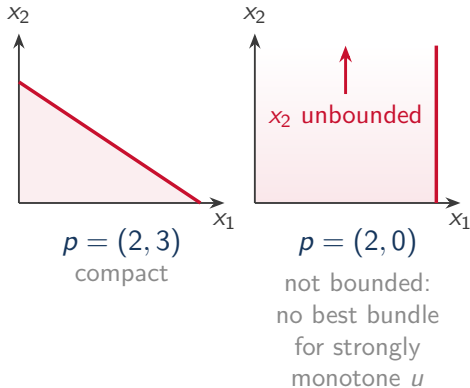
$$\max_{x \in \mathbb{R}_+^2} u(x) \text{ s.t. } p \cdot x \leq w \text{ (UMP)}$$

$\succsim$ rational, continuous, LNS; u continuous

$p \gg 0$, u continuous $\Rightarrow$ a solution exists

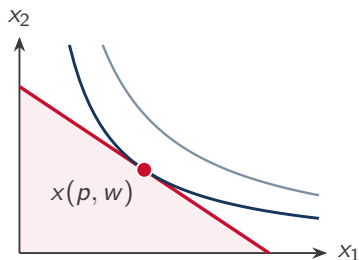
$\mathcal{B}_{p,w}$ closed and bounded ($x_\ell \leq w/p_\ell$): compact

MWG 3.D, Proposition 3.D.1 and its proof, pp. 50–51.



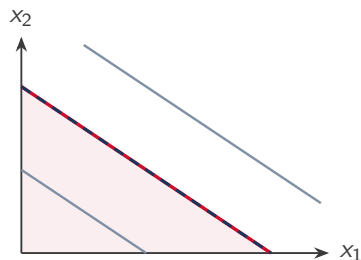
Walrasian demand: the best affordable bundles, one or many

$x(p, w)$ = the set of solutions of the UMP: **Walrasian demand**
a function when it is a single bundle at every (p, w) ; also called Marshallian demand



$$u = x_1 x_2$$

one bundle: (3, 2)



$$u = 2x_1 + 3x_2$$

every bundle on the line

MWG 3.D, Figure 3.D.1 and footnote 7.

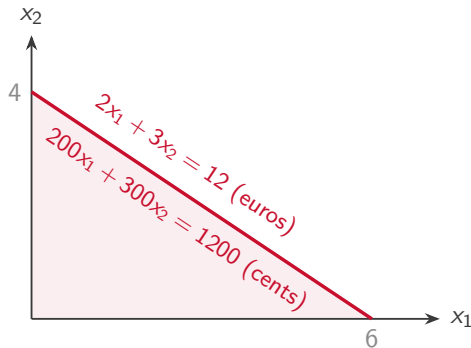
Scale all prices and wealth by $\alpha > 0$: demand does not move

$$\{x : \alpha p \cdot x \leq \alpha w\} = \{x : p \cdot x \leq w\}$$

so $x(\alpha p, \alpha w) = x(p, w)$ for all $\alpha > 0$

homogeneous of degree zero; no assumption on u : the set is the same

MWG Proposition 3.D.2(i) and its proof, p. 52; the same property is Definition 2.E.1.



Ann's raise buys about twice as much as Barbara's

Second year, relative to the prices p and salary w of the first:

	prices	salary	budget set
Ann	$\times 1$	$\times 1.02$	$\mathcal{B}_{p, 1.02w}$
Barbara	$\times 1.04$	$\times 1.05$	$\mathcal{B}_{1.04p, 1.05w} = \mathcal{B}_{p, 1.0096w}$

$\mathcal{B}_{p, 1.0096w} \subset \mathcal{B}_{p, 1.02w}$: Ann can afford everything Barbara can, and more.

Asked about feelings, people went by the nominal raise, not by what it buys: **money illusion**.

Shafir, Diamond & Tversky (1997), QJE 112(2), Problem 1; $1.05/1.04 \approx 1.0096$.

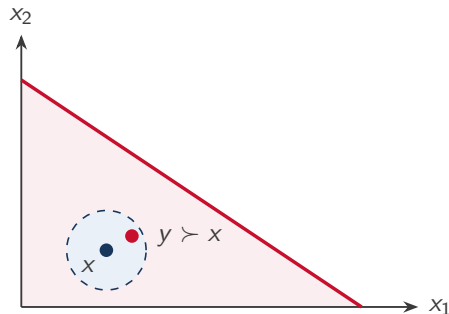
Walras' law: a locally non-satiated consumer spends all her wealth

$$p \cdot x = w \text{ for every } x \in x(p, w)$$

If $p \cdot x < w$:

a small disc around x is still affordable, and LNS puts a strictly better y in it

MWG Proposition 3.D.2(ii), Figure 3.D.2. LNS: class 2.



Strictly convex preferences: exactly one best bundle

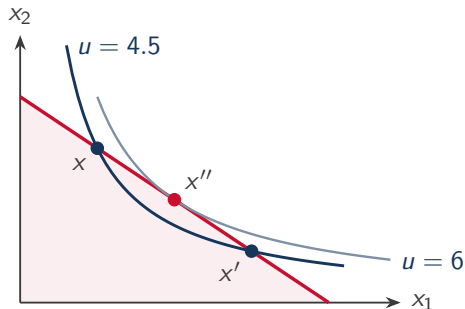
$\succsim$ convex $\Rightarrow x(p, w)$ is a convex set

$\succsim$ strictly convex $\Rightarrow x(p, w)$ is one bundle

two best bundles: their midpoint is affordable and strictly better

MWG Proposition 3.D.2(iii), Figure 3.D.3.

Convex $\succsim \iff$ quasiconcave u : class 3.



$$x'' = \frac{1}{2}x + \frac{1}{2}x': \text{cost } 12, u = 6 > 4.5$$

In pairs: could this be a Walrasian demand?

$x(p, w) =$	homogeneous of degree 0	Walras' law
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$$\left(\frac{w}{2p_1}, \frac{w}{2p_2} \right)$$

$$\left(\frac{w}{p_1 + p_2}, \frac{w}{p_1 + p_2} \right)$$

$$\left(\frac{w}{2p_1}, \frac{w}{3p_2} \right)$$

$$\left(\frac{w}{p_1 + 1}, \frac{w}{(p_1 + 1)p_2} \right)$$

Five minutes. Fail either test, and no LNS consumer has this demand.

In pairs: could this be a Walrasian demand?

$x(p, w) =$	homogeneous of degree 0	Walras' law	
$\left(\frac{w}{2p_1}, \frac{w}{2p_2}\right)$	✓	✓	Cobb–Douglas: slide 22
$\left(\frac{w}{p_1 + p_2}, \frac{w}{p_1 + p_2}\right)$	✓	✓	$u = \min\{x_1, x_2\}$
$\left(\frac{w}{2p_1}, \frac{w}{3p_2}\right)$	✓	✗	spends $5w/6$
$\left(\frac{w}{p_1 + 1}, \frac{w}{(p_1 + 1)p_2}\right)$	✗	✓	changes with the unit of money

Five minutes. Fail either test, and no LNS consumer has this demand.

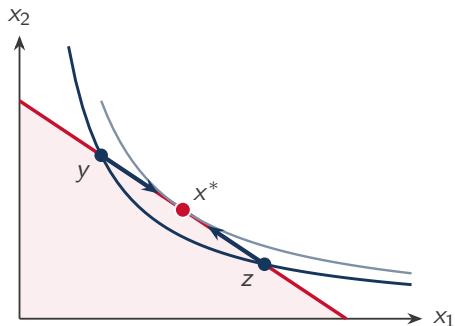
Break

Back at 11:00

First-order conditions

MWG 3.D

At an interior bundle with $MRS \neq p_1/p_2$, trading along the line helps



- at y : $MRS = 2 > 2/3$: more x_1
- at z : $MRS = 2/9 < 2/3$: more x_2
- at x^* : $MRS = 2/3 = p_1/p_2$

interior optimum: $MRS_{12}(x^*) = p_1/p_2$

$u = x_1x_2$, $p = (2, 3)$, $w = 12$; $MRS = x_2/x_1$ (class 2).

MWG 3.D, condition (3.D.5) and the argument after it, p. 54.

Cobb–Douglas: a constant share of wealth on each good

$u = x_1^\alpha x_2^{1-\alpha}$, $0 < \alpha < 1$; maximise $\alpha \ln x_1 + (1 - \alpha) \ln x_2$ instead same $\succsim$ (class 3)

$$\text{MRS} = \frac{\alpha x_2}{(1 - \alpha) x_1} = \frac{p_1}{p_2} \Rightarrow \frac{p_1 x_1}{p_2 x_2} = \frac{\alpha}{1 - \alpha}$$

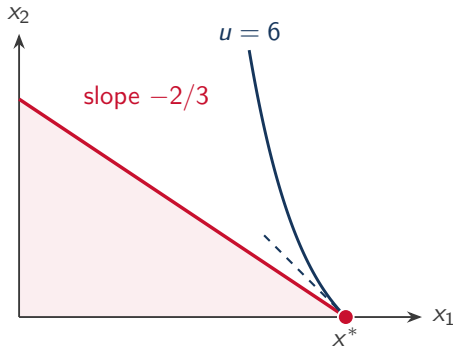
$$\text{with } p \cdot x = w: \quad x_1 = \frac{\alpha w}{p_1}, \quad x_2 = \frac{(1 - \alpha) w}{p_2}$$

$$\alpha = \frac{1}{2}, \quad p = (2, 3), \quad w = 12: \quad x = (3, 2) \quad \text{slide 21}$$

A share α of wealth goes to good 1, whatever the prices.

MWG Example 3.D.1 (there $u = k x_1^\alpha x_2^{1-\alpha}$), pp. 55–56.

The same recipe can ask for a negative quantity



$$u = x_1 + \ln(1 + x_2)$$

class 3, slide 27; $p = (2, 3)$, $w = 12$

$$\text{MRS} = 1 + x_2 = \frac{2}{3} \Rightarrow x_2 = -\frac{1}{3}$$

best affordable: $x^* = (6, 0)$

there $\text{MRS} = 1 > 2/3$: the curve is steeper than the line

Tangency is a condition for interior bundles only.

MWG Figure 3.D.4(b): a boundary solution.

Kuhn–Tucker: one set of conditions for interior and corner

u continuously differentiable, $x^* \in x(p, w) \Rightarrow$ there is $\lambda \geq 0$ with

$$\frac{\partial u(x^*)}{\partial x_\ell} \leq \lambda p_\ell \text{ for each good } \ell, \text{ with equality if } x_\ell^* > 0 \quad (3.D.1)$$

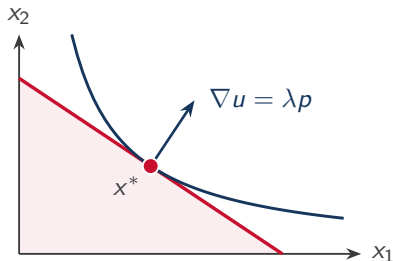
in vectors: $\nabla u(x^*) \leq \lambda p$ and $x^* \cdot [\nabla u(x^*) - \lambda p] = 0$ (3.D.2), (3.D.3)

sufficient too, for x^* with $p \cdot x^* = w$,

if u is quasiconcave, monotone, $\nabla u(x) \neq 0$ for all x p. 55

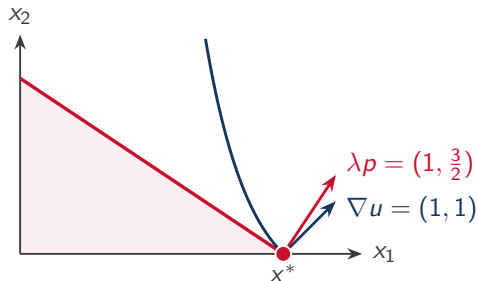
MWG 3.D, p. 53; the constraint qualification always holds in the UMP (footnote 8).

At a corner the gradient need not point along p



$u = x_1 x_2$, **interior**

$$\nabla u(x^*) = (2, 3) = \lambda p, \quad \lambda = 1$$



$u = x_1 + \ln(1 + x_2)$, **corner**

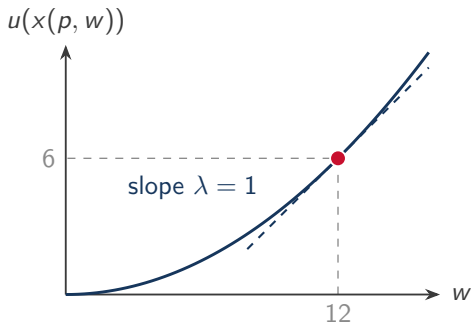
$$x_2^* = 0: \quad \partial u / \partial x_2 = 1 < \lambda p_2 = \frac{3}{2}, \quad \lambda = \frac{1}{2}$$

$\nabla u(x^*) \leq \lambda p$, with equality in every good that is bought.

MWG Figure 3.D.4.

λ is the marginal utility of wealth, and it is cardinal

u	λ at $x^* = (3, 2)$	$\frac{d}{dw} u(x(p, w))$ at $w = 12$
$x_1 x_2$	1	$w/12 = 1$
$\frac{1}{2} \ln x_1 + \frac{1}{2} \ln x_2$	$1/12$	$1/w = 1/12$



$$u = x_1 x_2: u(x(p, w)) = w^2/24$$

Same $\succsim$, same bundle, λ rescaled by $f'(u)$: like marginal utility (class 3).

MWG 3.D, pp. 54–55; $p = (2, 3)$.

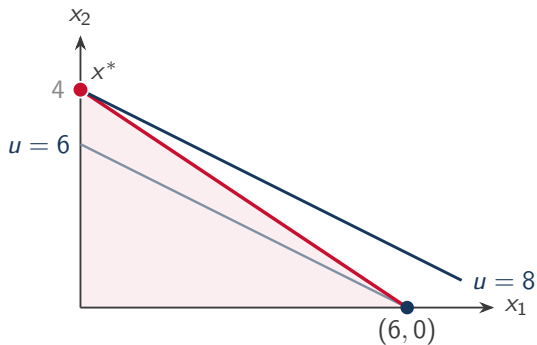
Perfect substitutes: all on the good with more utility per euro

$u = ax_1 + bx_2$: compare $\frac{a}{p_1}$ and $\frac{b}{p_2}$

$\frac{a}{p_1} > \frac{b}{p_2}$: $x = (w/p_1, 0)$

$\frac{a}{p_1} < \frac{b}{p_2}$: $x = (0, w/p_2)$

equal: the whole budget line slide 13



Here $u = x_1 + 2x_2$: $\frac{1}{2} < \frac{2}{3}$, so $x^* = (0, 4)$; Kuhn–Tucker: $\lambda = \frac{2}{3}$, and $1 < \frac{4}{3}$ for good 1.

In pairs: Kuhn–Tucker for $u = x_1 + \ln(1 + x_2)$

Find $x(p, w)$ and λ ; which condition is a strict inequality?

p	w	$x(p, w)$	λ	strict inequality
(3, 1)	12			
(3, 1)	1			
(1, 1)	4			

Six minutes. Try interior first; a negative quantity means: set it to zero, check the inequality.

In pairs: Kuhn–Tucker for $u = x_1 + \ln(1 + x_2)$

Find $x(p, w)$ and λ ; which condition is a strict inequality?

p	w	$x(p, w)$	λ	strict inequality
(3, 1)	12	(10/3, 2)	1/3	none: interior
(3, 1)	1	(0, 1)	1/2	good 1: $1 < 3/2$
(1, 1)	4	(4, 0)	1	none, although $x_2 = 0$

Six minutes. Try interior first; a negative quantity means: set it to zero, check the inequality.

Row 3: a zero and an equality at once: “equality if $x_\ell > 0$ ” means if, not only if.

CES demand: the lower σ , the more balanced the bundle

$$u = \left(\frac{1}{2}x_1^\rho + \frac{1}{2}x_2^\rho \right)^{1/\rho}, \quad \sigma = \frac{1}{1-\rho}$$

class 3, slide 26

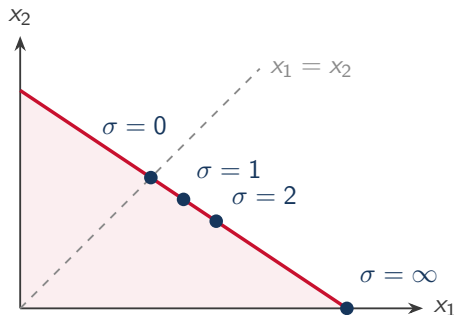
$$x_\ell(p, w) = \frac{w p_\ell^{-\sigma}}{p_1^{1-\sigma} + p_2^{1-\sigma}}$$

$\sigma = 1$: $w/(2p_\ell)$, Cobb–Douglas

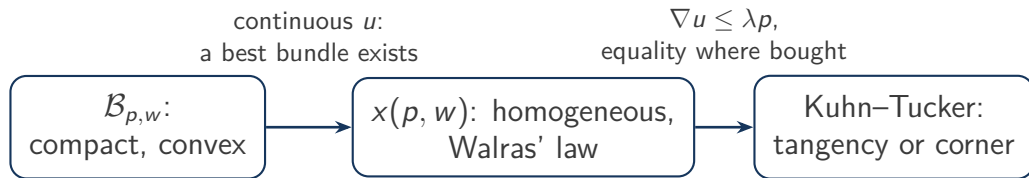
$\sigma = 0$: $w/(p_1 + p_2)$, Leontief

$\sigma \rightarrow \infty$: all on the cheaper good

$\rho = (2, 3)$, $w = 12$; interior for $\rho < 1$, the perfect-substitutes corner at $\rho = 1$.



Today in one line



Tuesday: indirect utility; wealth and price effects; Roy's identity.

Problem set 1: due at the start of class Wed 14 Oct, one copy per group.

Read MWG 2.E, 3.D and 3.G (or Muñoz-Garcia ch. 2) for Tuesday 13 October.